Maximal Counts in the Stopped Occupancy Problem

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Summary

In this talk we take a point process approach to revise and refine some extreme-value results on the classic balls-in-boxes occupancy problem, with a view to understanding both static and dynamical aspects.

This involves:

- Continuous-time embedding via single and bi-Poissonisation.
- Asymptotics of stopped maximal counts.
- Oscillatory phenomena.
- Connection to infinite tandem service queues.

(a version appears in de Moivre's 1718 Doctrine of Chances)

Coupons of n types are found in cereal packs. Let τ_n denote the total number of packs required to observe at least m representatives of every type.

Well known asymptotics:

- For a single complete set ($m = 1$):

$$\mathbb{E}[\tau_n] = n(\log n + \gamma) + \frac{1}{2} + o(1) \quad \text{as } n \to \infty$$

- For the generalised Double Dixie Cup problem ($m \geq 2$ full sets):

$$\mathbb{E}[\tau_n] = n\big(\log n + (m-1) \log\log n + c_m + o(1)\big)$$

The Classical Occupancy Scheme

packs $\to$ balls, coupon types $\to$ boxes

N balls allocated independently to n boxes, with same probability $1/n$ for landing in each box.

- **Kolchin, Sevastyanov and Chistyakov 1978 book:** focused on multiplicity counts tracking boxes with exactly $r = 0, 1, \ldots$ balls, in various asymptotic regimes for n, N.
- **Ivchenko** in a series of papers 1973–1996 studied configurations at random times τ_n.

Problem Statement

Let balls be sequentially allocated into n boxes until random stage τ_n, when there remain exactly ℓ boxes containing at most m balls each. Parameters $m \geq 0, \ell \geq 0$ are fixed throughout, with $\ell = 0, m = 1$ corresponding to the CCP case.

Objective: to understand the large-n behaviour of a few top components of the order statistics vector at stopping,

$$\boldsymbol{M_n}(\boldsymbol{\tau_n}) := \big(M_{n,1}(\tau_n) \geq M_{n,2}(\tau_n) \geq \cdots \geq M_{n,n}(\tau_n)\big),$$

and, more generally at times when the number of balls thrown is of the order $n \log n$ as in the CCP.

Poissonisation

- Balls are thrown in boxes at times of independent Poisson processes $(\Pi_k(t), t \geq 0)$ of unit rate.
- By time $t \asymp \log n$, about $nt \asymp n \log n$ balls are thrown.
- The allocation of balls (in unlabeled) boxes can be encoded into ordered occupancy counts or multiplicities

$$\mu_{n,r}(t) := \#\{k \leq n : \Pi_k(t) = r\} = \#\{k \leq n : M_{n,k}(t) = r\}.$$

  In particular, $M_{n,1}(t)$ is the largest r with $\mu_{n,r}(t) \neq 0$.
- Jointly, the $\mu_{n,r}(t)$'s have a multinomial distribution, for fixed t.

Bi-Poissonisation

Replaces fixed number of boxes n by random Poisson(n).

- Multiplicities (further denoted same way as for fixed n) $\mu_{n,r}(t)$ for distinct occupancy levels r become independent for any fixed time t.
- With distribution

$$\mu_{n,r}(t) \stackrel{d}{=} \text{Poisson}\big(n \cdot p_r(t)\big), \quad \text{where } p_r(t) := \frac{e^{-t} t^r}{r!}.$$

- The (rough) range of interest is $t \sim \log n$ and $r \sim e \log n$.
- For each fixed K, the total variation distance between vectors of K maximal order statistics in poissonised and bi-poissonised models decays as a power of n. We make therefore no difference between the schemes in the sequel.

Exponential Poisson Point Processes

Let Ξ be an inhomogeneous Poisson point process (PPP) on $\mathbb{R}$ with intensity measure $e^{-x}\mathrm{d}x$. The atoms of Ξ form a strictly decreasing sequence $\xi_1 > \xi_2 > \cdots \to -\infty$.

- The maximal point ξ_1 has the standard double-exponential (Gumbel) distribution:

$$\mathbb{P}[\xi_1 \leq x] = \exp(-e^{-x}), \quad x \in \mathbb{R}.$$

- The k-th ordered atom follows a generalised Gumbel distribution

$$\xi_k \stackrel{d}{=} \mathrm{Gumbel}(k) \quad \text{with density} \quad e^{-x} p_{k-1}(e^{-x}).$$

Random shift Invariance

Let $\Xi_b := \Xi + b$ denote the exponential PPP translated by $b \in \mathbb{R}$.

- Poisson shift T_h acts on a point configuration by adding to each atom an independent Poisson(h) random variable.
- The action of T_h on Ξ corresponds to a deterministic translation:

$$T_h \circ \Xi \overset{d}{=} \Xi_{(e-1)h}$$

- Letting $h \geq 0$ vary yields a *Poisson flow*, in which each particle independently moves to the right by unit jumps at unit rate, with the point configuration for each h being a PPP.

For $\Xi_b = \Xi + b$ exponential PPP translated by $b \in \mathbb{R}$:

The lattice PPP analogue $\Xi_b^{\uparrow}$ is obtained by upper rounding:

$$\Xi_b^{\uparrow} := \{\lceil \xi_i + b \rceil\} \subset \mathbb{Z}$$

- Cumulative distribution functions match exactly at integer lattice endpoints:

$$\mathbb{P}[\lceil \xi_i + b \rceil \leq r] = \mathbb{P}[\xi_i + b \leq r] = P_{i-1}(e^{-r+b}), \quad r \in \mathbb{Z},$$

  where $P_r(t) := p_0(t) + \cdots + p_r(t)$.

- The mean number of atoms at integer locations is a two-sided geometric sequence:

$$\mathbb{E}[\Xi_b^{\uparrow}(\{r\})] = e^{-r+b}(e-1).$$

Let $\alpha_n := L + m \log L - \log m!\,,\ L := \log n$.

Proposition

$$\tau_n - \alpha_n \xrightarrow{d} \tau \quad \text{as } n \to \infty,$$

where $\tau \overset{d}{=}$ *Gumbel*$(\ell + 1)$ *has density*

$$\mathbb{P}[\tau \in \mathrm{d}s] = \frac{1}{\ell!} \exp\left(-(\ell+1)s - e^{-s}\right) \mathrm{d}s, \quad s \in \mathbb{R}.$$

- For times $t = \alpha_n + O(1)$ the multiplicities are

$$\mu_{n,m}(t) = O_p(1) \quad \text{(Bounded in Probability)}$$
$$\mu_{n,r}(t) = o_p(1) \xrightarrow{p} 0 \quad \text{for all lower counts } r < m$$
$$\mu_{n,r}(t) \xrightarrow{p} \infty \quad \text{for all higher counts } r > m$$

- $a_n := eL + \left((e-1)m - \frac{1}{2}\right)\log L - \log\left(\frac{(e-1)m!^{e-1}}{\sqrt{2\pi e}}\right)$,
- $b_n = \lfloor a_n \rfloor, \quad c_n = \{a_n\} \in [0,1)$.

Proposition

Let $t = L + m\log L + u$ and $r = eL + ((e-1)m - 1/2)\log L + v$. Then, as $L \to \infty$, for $\overline{P}_r(t) := 1 - P_r(t)$

$$\overline{P}_r(t) \sim \frac{1}{e-1} p_r(t)$$

$$\log p_r(t) = -L + (e-1)u - v - \log\sqrt{2\pi e} - \frac{(eu-v)^2}{2eL} + O\left(\frac{\log L}{L^{1/2}}\right)$$

These approximations hold uniformly in u, v provided $|eu - v| = O(L^{1/2})$, as $L \to \infty$.

Theorem (Ivchenko+GJM)

Let $M_n := M_{n,1}(\tau_n)$ be the maximum occupancy count at time τ_n. For $k \in \mathbb{Z}$, as $n \to \infty$:

$$\mathbb{P}[M_n - b_n \leq k] = \int_{-\infty}^{\infty} p_\ell(e^{-s}) e^{-s} \exp\left(-e^{(e-1)s + c_n - k}\right) \mathrm{d}s + o(1).$$

Equivalently,

$$\lim_{n \to \infty} d_{TV}(M_n - b_n, Z_n) = 0$$

where the approximating random variable is defined by:

$$Z_n := \lceil \xi_1 + (e-1)\tau + c_n \rceil,$$

with $\xi_1 \overset{d}{=} \mathrm{Gumbel}(1), \tau \overset{d}{=} \mathrm{Gumbel}(\ell + 1)$ being independent. A weak convergence does not hold due to the about log-periodic c_n.

Derived point processes

- The maximal counts are captured by the lattice PPP

$$M_n^s := \sum_{k=-\infty}^{\infty} \mu_{n,b_n+k}(\alpha_n + s)\delta_k$$

- Let $R_{n,r}$ be the point process of times when a box receives its rth ball (r-record times). These processes are non-homogeneous Poisson with intensity measure $np_{r-1}(t)\mathrm{d}t$.

Approximating by the exponential PPP

Theorem

Let $r_n \sim \beta \log L$ for any $0 < \beta < 1/2$. Then locally uniformly in $s \in \mathbb{R}$:

$$d_{TV}\left(M_n^s\big|_{[-r_n,\infty]},\ \Xi^{\uparrow}_{(e-1)s+c_n}\big|_{[-r_n,\infty]}\right) \to 0$$

Corollary: For every $K \in \mathbb{Z}_{>0}$, as $n \to \infty$ locally uniformly in $s \in \mathbb{R}$:

$$d_{\mathrm{TV}}\left(\left(M_{n,i}(\alpha_n + s) - b_n\right)_{i=1}^{K},\ \left(\lceil \xi_i + (e-1)s + c_n \rceil\right)_{i=1}^{K}\right) \to 0$$

where $\xi_1 > \xi_2 > \dots$ are the ordered atoms of the exponential PPP Ξ.

Steps of the proof:

1. Restrict process comparisons to the window $[-r_n, L^{1/4}]$, where $L = \log n$.
2. Use the asymptotic expansions of Poisson probabilities.
3. Verify that the ℓ_1-distance between the underlying intensity measures vanishes.

Extreme Coordinate Decoupling

Proposition

Let $t_0 = \frac{2}{3}L$ and $t_1 = L + (K-1)\log L$. The process of the K-minimal counts and the process of the K-maximal counts are asymptotically independent over the temporal window $[t_0, t_1]$.

Theorem

The joint distribution of the upper order statistics at the stopping time τ_n satisfies:

$$\lim_{n\to\infty} d_{TV}\left(\left(M_{n,i}(\tau_n) - b_n\right)_{i=1}^{K},\ \left(\lceil \xi_i + (e-1)\tau + c_n \rceil\right)_{i=1}^{K}\right) = 0$$

where the random stopping time $\tau \stackrel{d}{=} \mathrm{Gumbel}(\ell+1)$ is independent of the atoms $\xi_i \stackrel{d}{=} \mathrm{Gumbel}(i)$.

Step 1: Conditional Independence

- Given $\tau_n = t$, the occupancy counts $\Pi'_1, \ldots, \Pi'_{n-\ell-1}$ of the remaining $n-\ell-1$ boxes are independent and follow a truncated Poisson distribution on $\{m+1, m+2, \ldots\}$.

Step 2: Non-lattice Extension

- Let $E_1, \ldots, E_{n-\ell-1}$ be independent truncated exponential variables on $[0,1)$ with density $f_E(x) = e^{-x}/(1-e^{-1})$. Define the continuous interpolated variables

$$\tilde{\Pi}'_i := \Pi'_i + E_i, \quad i = 1, \ldots, n-\ell-1$$

Step 3: Large-t Point Process Approximation

- For large time horizons $t \asymp \log n$, the properly shifted and scaled interpolated variables $\hat{\Pi}_i$ satisfy $\mathbb{P}[\hat{\Pi}_i > x] = \frac{1}{n}(e^{-x} + o(1))$.
- The induced point process $\hat{\Xi}'_n$ approaches the exponential PPP Ξ with intensity $e^{-x}\mathrm{d}x$.

Step 4: joint convergence

- A general theorem establishes the joint weak convergence:

$$\left(\hat{\Xi}'_n(\cdot),\, \tau_n - \alpha_n\right) \xrightarrow{d} (\Xi(\cdot),\, \tau) \quad \text{as } n \to \infty$$

- By the Continuous Mapping Theorem (Composition Theorem)

$$\{\Pi'_i(\tau_n) + E_i - a_n - 1\}_{i=1}^{n-\ell-1} \xrightarrow{d} \Xi + (e-1)\tau.$$

Step 5: back to discrete variables

- Assuming along subsequence $c_n \to c$:

$$M_{n,i} - b_n = \lfloor \tilde{\Pi}_{(i)} - b_n \rfloor \xrightarrow{d} \lfloor \xi_i + (e-1)\tau + 1 + c \rfloor,$$

- where the RHS is desired

$$\lfloor \xi_i + (e-1)\tau + 1 + c \rfloor = \lceil \xi_i + (e-1)\tau + c \rceil,$$

with $\xi_1 > \xi_2 > \dots$ being the atoms of Ξ, and $\tau \sim \text{Gumbel}(\ell + 1)$ independent of Ξ.

Proposition

Let $X_{n,1} \geq \cdots \geq X_{n,n}$ be the order statistics of a sample of size n from a distribution F_n. Choose a sequence x_n such that

$$c_- n^{-1} \leq \overline{F}_n(x_n) \leq c_+ n^{-1}.$$

If the hazard ratios satisfy $\overline{F}_n(x+y)/\overline{F}_n(x) \leq Ce^{-cy}$ for $\theta x_n \leq x \leq x_n$, then for $y \geq 0$:

$$\mathbb{P}[X_{n,i} - x_n > y] \leq (c_+ C)e^{-cy},$$

$$\mathbb{P}[X_{n,i} - x_n \leq -y] \leq C_2 e^{-c_2 e^{c(1-\theta)y}}$$

The result is aplicable e.g. to Gamma and Poisson distributions.

Approximating the Moments

Theorem

For $i \geq 1$ and $k \geq 1$, as $n \to \infty$:

$$\mathbb{E}\left[(M_{n,i}(\tau_n) - b_n)^k\right] = \mathbb{E}\left[\lceil \xi_i + (e-1)\tau + c_n \rceil^k\right] + o(1)$$

Using Janson (2006), the expectation expands into the Fourier series

$$\mathbb{E}[M_{n,j}] = a_n + \gamma e - H_{j-1} - (e-1)H_\ell + \frac{1}{2} + \sum_{k \neq 0} \chi_k e^{2\pi \mathrm{i} k c_n} + o(1)$$

where γ is Euler's constant, H_r are harmonic numbers, and the coefficients are:

$$\chi_k = \frac{\Gamma(j - 2\pi \mathrm{i} k)\Gamma(\ell + 1 - 2\pi(e-1)\mathrm{i} k)}{(j-1)!\ell! \cdot 2\pi \mathrm{i} k}$$

Oscillation Amplitude:
The product of the two complex Gamma functions forces an extremely fast decay on χ_k. For $j = 1, \ell = 0$, the total oscillation amplitude is bounded below $4 \cdot 10^{-11}$, making the fluctuations negligible.

Let Q_n be the number of boxes achieving the maximum stopped count:

$$Q_n := \min\{i : M_{n,i} > M_{n,i+1}\} = \mu_{n,M_n}(\tau_n)$$

Theorem

As $n \to \infty$, for $j \in \mathbb{Z}_{>0}$

$$\mathbb{P}[Q_n = j] = \chi_j(c_n) + o(1)$$

where $\chi_j(u)$ is a continuous, 1-periodic function given by

$$\chi_j(u) = \frac{(1-e^{-1})^j}{j}\left(1 + \sum_{k\in\mathbb{Z}\setminus\{0\}} \frac{\Gamma(j-2\pi\mathrm{i}k)\Gamma(\ell+1-2\pi(e-1)\mathrm{i}k)}{(j-1)!\ell!}e^{2\pi\mathrm{i}ku}\right)$$

The mean value of $\chi_j(u)$ matches the standard log-series distribution. While the unstopped maximum ties probability is of the order 10^{-4}, convolution with the continuous density of τ suppresses the periodic fluctuations down to 10^{-10}.

In the bi-poissonised framework, let $R_{n,r}$ denote the point process on $\mathbb{R}_+$ formed by the times when a box receives its r-th ball (r-records). The mean measure is given by:

$$\mathbb{E}[R_{n,r}(\mathrm{d}t)] = np_{r-1}(t)\mathrm{d}t = n\frac{e^{-t}t^{r-1}}{(r-1)!}\mathrm{d}t$$

Let

$$\alpha_{n,r} := L + (r-1)\log L - \log(r-1)! \quad \text{where } L := \log n.$$

Theorem

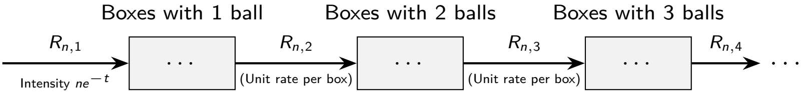
$$R_{n,r}(\alpha_{n,r} + \cdot) \xrightarrow{d} \Xi \quad \textit{as } n \to \infty$$

where Ξ is the exponential PPP with intensity measure $e^{-s}\mathrm{d}s$.

The $r = 1$ instance ($\alpha_{n,1} = \log n$) is equivalent to Sevastyanov's (1967) result on the decay process of empty boxes.

Infinite-server queues in tandem

The bi-poissonised occupancy process is representable as an infinite series of independent $M_t/M/\infty$ queues in tandem.



- **Input**: the input process $R_{n,1}$ corresponds to the times when empty boxes receive their first ball, which for large n approximates Sevastyanov's empty boxes decay process.
- **Unit rate transitions**: each ball moves at unit rate to the next category of boxes.
- **Internal flows**: the departure stream moving from the r-th queue to the $(r+1)$-th queue, $R_{n,r+1}$, has mean intensity measure $np_r(t)\mathrm{d}t$.
- **Independence properties**: by the standard properties of open Poisson networks (Kelly, 2011), the queue contents are independent at any fixed time t, validating the bi-poissonised independence of multiplicities.